



From Record to Finding: Machine-Checkable Evidentiary Adequacy for Agentic AI Oversight under EU Law

Abstract: Under EU law a supervisor of an agentic AI system must establish specific findings of fact: whether protected data left a controlled environment, whether a human held the authority and the window to intervene, whether an information barrier held, whether a delegated authority was valid when used. This paper asks a prior, tractable question: under what conditions can a runtime record *answer* such a finding at all, and how can that condition be checked by machine? We develop an evidentiary-adequacy criterion: a representation answers such a determination only if it carries a *legal typing* mapping recorded events to the operative category, and the *relation*—provenance, authority, or temporal validity—on which the determination’s truth depends. We then operationalise the criterion as a legal knowledge-representation artefact. Its two features are realised by reuse of PROV-O and the Data Privacy Vocabulary over a supervisory-evidence ontology, and rendered machine-checkable as SHACL shapes, among them a shape requiring typed validity intervals on delegated authority and a SPARQL constraint operationalising the Court of Justice’s reasoning in *SCHUFA* (C-634/21). In a worked validation an under-typed record fails conformance and a typed relational record conforms. Relating conformance to the AI Act’s Article 12 / Article 14 split, we show that a record satisfying the letter of logging can still fail, by machine-checkable test, to evidence that effective oversight was possible. The contribution is a formal criterion, its executable rules-as-code operationalisation, a gap analysis against governance frameworks and accountability ontologies, and a pre-registered evaluation.

Keywords: computational law, legal knowledge representation, rules as code, SHACL, EU AI Act

1. Introduction

An agentic AI system is software that uses a model to decide what to do and a harness to do it. One property distinguishes it from ordinary software: its execution path, the sequence of steps it takes, is selected at run time and cannot be enumerated in advance, and that selection can be steered by the content the agent reads. A conventional service fixes its control flow at design time; an agent composes its control flow as it runs.

The dominant response in enterprise governance has been to extend the instruments built for conventional software: the point-in-time audit, the conformity checklist, the tamper-evident log. These produce records. Under EU law, however, a supervisor is required not merely to hold records but to establish *findings of fact* from them: that protected data did or did not cross a boundary, that a barrier did or did not hold, that a human could or could not have intervened. This paper asks what a record must contain for a supervisor—or a supervisory information system—to read a legally operative fact from it at all, and it treats that question computationally: as a question about the structure a machine-readable evidence graph must have for the finding to be recoverable, and checkable, by a decision procedure over the graph.

1.1. The contribution, stated up front

This paper makes four contributions, and states them in the vocabulary of computational law rather than of governance theory.

1. A *formal criterion* (legal knowledge representation). We define an *evidentiary-adequacy criterion*: for a class of legal

determinations that are findings of fact about specific events and their relations, a runtime record can answer the determination only if it represents, in explicit or functionally equivalent form, (i) a *typing* that maps recorded events to the operative legal category, and (ii) the *relation*—provenance, authority, or temporal validity—on which the determination’s truth depends. Necessity is shown by construction across the class; the two-feature set is shown minimal; adequacy is bounded to the class. Framed in the vocabulary of the analysis of legal proof [1], the two features are the minimum representational substrate from which an evidentiary argument connecting a record to a legally material *probandum* can be constructed—not the argument itself, and not a measure of its probative force.

2. An *executable operationalisation* (rules as code). We realise the criterion in machine-readable artefacts: a supervisory-evidence ontology in OWL/RDF that reuses PROV-O for the relation and the Data Privacy Vocabulary (DPV) for the typing, and a set of SHACL shapes that render the criterion’s conditions machine-checkable. Two shapes are of independent computational-law interest—one requires a typed validity interval and authority scope on delegated authority, and one encodes, as a SPARQL constraint, the Court of Justice’s reasoning in *SCHUFA* (C-634/21) on when an upstream score must itself be treated as an automated decision.
3. A *worked legal application and gap analysis*. We instantiate the criterion against selected obligations of the General Data Protection Regulation (GDPR)¹ and the EU Artificial

¹Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard

Intelligence Act,² give a worked validation in which an under-typed record fails SHACL conformance and a typed relational record conforms, and show by framework-by-framework analysis that no governance framework in current use, and no existing accountability ontology, produces by itself a record that satisfies the criterion.

4. *A pre-registered evaluation.* We give a between-groups experiment testing whether expert panels can resolve the in-scope determinations from bare logs, provenance-only records, or typed relational representations, and a technical throughput benchmark for the SHACL validation, so that the criterion is exposed to falsification rather than asserted.

1.2. Why this is a computational-law contribution

The paper's object is not a theory of oversight but a *representation* problem with a legal target: it specifies the structure a compliance-checkable evidence record must have, encodes selected statutory obligations and case-law distinctions as machine-checkable constraints, and demonstrates the resulting rules-as-code on worked data. It sits in an established lineage. Rules-as-code work renders statutory provisions as executable artefacts whose behaviour can be tested against the text they encode [7]; semantic-web legal modelling has produced vocabularies for data-protection obligations [4] and for the AI Act's risk and high-risk classifications [6], and a systematic analysis of which ontologies and policy languages can carry GDPR's informational items [5]. This paper's target within that lineage is narrower and operational: a machine-readable fabric for supervisory *evidence* about agentic AI under European law, and a decidable test for when such evidence can and cannot ground a legal finding.

1.3. Related work

Four bodies of work bear directly on the problem, and none of them settles it.

Accountability and answerability. Accountability in AI has been analysed as a relation of answerability between an agent and a forum, decomposable into context, range, agent, forum, standards, process, and implications [9]. That analysis fixes *who* must answer to *whom*; the criterion developed here concerns the prior question of whether the record can answer at all. Reviewability proposes to decompose automated decision-making into technical and organisational elements and to specify contextually appropriate record-keeping across them [10], and traceability has been advanced as the principle that makes transparency serve accountability [12]. Both establish that records must be designed for review; neither states a decidable condition on record content, which is what this paper adds. The accountability problem is further complicated where systems are assembled across supply chains [11], and where the agent's own actions are unobservable to those who must oversee them [13].

Auditing and access. Work on AI auditing establishes that audit quality is a function of the access granted: black-box query access is insufficient for rigorous audits, and stronger conclusions require access to internals [14]. The conformity-assessment and

post-market-monitoring machinery of the AI Act has been mapped onto the auditing literature [15]. These results concern access to the *system*; the criterion here concerns the structure of the *record* the system emits, which remains under-specified even where access is total.

Human oversight. Empirical and doctrinal work has shown that oversight requirements are frequently unsatisfiable in practice [16], that the Act's oversight duty is underdetermined as to what is overseen, when, and by whom [17], and that effectiveness in oversight is an interdisciplinary construct rather than a checkbox [18]. This literature asks whether the human can oversee; the present paper asks the logically prior question of whether the record permits the operative fact to be established by anyone, human or machine.

Accountability ontologies. Closest in method are semantic artefacts for accountability: the System Accountability Ontology and its RAInS extension model accountability plans and their traces for machine-learning systems [8], and AI Act vocabularies model risk and high-risk classification [6]. These model the *governance process*—plans, roles, risks, obligations—rather than the runtime evidence graph from which a finding of fact about a particular execution is recovered, and they do not state, or check, a necessity condition on that graph. Section 4 makes the comparison explicit.

1.4. What this paper does not claim

It does not claim that legal classification can be automated; a SHACL-conformant graph is a claim of structural completeness against a declared profile, not a proof of compliance, and human legal classification remains necessary. It does not claim that a particular ontology is mandated by EU law. It does not claim the criterion is sufficient for good oversight; sufficiency is out of scope and shown to fail. It does not claim novelty for the cybernetic observation, used only as convergent support, that an overseer must model the system it oversees.

2. The Evidentiary-Adequacy Criterion

This section states the contribution before any operationalisation. The criterion stands on the argument given here; Sections 3 and 5 corroborate it computationally and theoretically, but it does not depend on them.

2.1. Determinations, evidence, answerability

Let a *determination* D be a legally operative finding of fact that a supervisory system must be able to establish about an agentic system's behaviour. Let an *evidence representation* E be the record, in whatever form, from which the supervisory system attempts to establish D . Let an *answering procedure* be a decision procedure f that takes E and returns a truth value for D .

Definition 1 (answerability). E *answers* D when there exists a decision procedure f such that $f(E)$ recovers the truth value of D , soundly, for the cases in the determination's scope. E *fails to answer* D when no such procedure exists over E , that is, when the truth value of D is not a function of the content of E .

The term is chosen advisedly. Where accountability is analysed as answerability—an agent's obligation to answer to a forum against a standard [9]—this definition isolates the representational precondition of that obligation: the agent can answer only about what its record makes recoverable. This is deliberately modest. It says nothing about how E is produced or whether oversight as a whole succeeds. It isolates one question: is the fact recoverable from

to the processing of personal data (General Data Protection Regulation), OJ L 119, 4.5.2016, pp. 1–88; Articles 5(1)(b), 6(1), 15 and 22.

²Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), OJ L, 2024/1689, 12.7.2024; Articles 12, 14, 19, 26, 72 and 113.

the record, or is it not present in the record to be recovered? That question is exactly the one a SHACL validator, a SPARQL query, or any decision procedure over a graph must be able to settle, which is why the criterion admits a machine-checkable form.

2.2. Which determinations: the selection rule

Selection rule. A determination is *in scope* when it is a binary finding of fact about specific events and their relations: a yes-or-no question whose truth value is fixed by what specific events occurred, what authority they ran under, and how they relate to one another. A determination is *out of scope* when it is an evaluative or statistical property over a distribution of behaviours rather than a finding about particular events.

The four determinations used throughout are in scope by this rule: whether protected data crossed a boundary, whether a human held authority and a usable window to intervene, whether an information barrier held, whether a delegated authority was valid when used. Fairness, robustness, non-discrimination, explainability, and transparency are out of scope: they are properties over distributions of outcomes, and they raise representational questions the criterion does not address. Excluding them is not a weakness; it is the boundary that makes the criterion precise.

2.3. The criterion

Criterion (evidentiary adequacy). Let D be a determination in scope, of the form “did event-type X , of legal category C , stand in relation R to event-type Y ?” Then E answers D only if E represents, explicitly or in functionally equivalent form, both:

1. a *typing* that maps the recorded events to the legal category C ; and
2. the *relation* R —usually provenance, derivation, authority, or temporal validity—on which the truth of D depends.

A representation carrying neither the typing nor the relation can, at most, detect that something occurred or that something looks anomalous. It cannot establish D as a finding of fact. The criterion is a necessary condition; it is not claimed to be sufficient (Section 9).

2.4. Relation to the analysis of legal proof

The analysis of legal evidence distinguishes the *ultimate* and *penultimate probanda* specified by substantive law—the legal conclusion to be established, and the elements that must be established for it—from the evidentiary propositions through which those matters are proved; on that account evidence is relevant not in isolation but through a defensible inferential chain connecting it to a legally material proposition [1]. The criterion developed here concerns a *prior* representational condition for constructing such a chain. The legal typing permits a recorded event to be mapped to a legally material category; the relational structure preserves the provenance, authority, and temporal connections on which that mapping may depend. Neither feature establishes probative force or satisfies a standard of proof. Without them, however, the evidentiary argument cannot be formulated from the runtime record at all. The criterion thus identifies the substrate from which a legal evidentiary argument can be built; it is not that argument, and Section 9 states what it leaves to it.

2.5. Necessity by construction

Necessity is shown, determination by determination, by exhibiting a bare action log and demonstrating that the truth value is not a function of the log’s content. A bare action log records, for each action, that it occurred, when, and that it was permitted, with no typing of data or channel and no representation of derivation or authority state. Table 1 states the construction.

2.6. Minimality and bounded completeness

The two-feature set is minimal. Drop the typing: for the first determination the log may record, with provenance, that a send crossed an external boundary, but cannot say the content was of a protected class, so “protected data left the environment” is not recoverable. Drop the relation: for the third determination the log may type each action but cannot represent that the send was derived from the read, so “the barrier held” is not recoverable. Neither feature is redundant.

Completeness is claimed only for the class defined by the selection rule, and only where “relation” is read broadly enough to carry provenance, authority, and the temporal ordering on which validity depends. Two of the four determinations turn on time and authority state, not derivation alone; the paper reads “relation” to include them, explicitly. Universal completeness is overreach the paper declines, for the same reason it declines an if-and-only-if theorem in Section 9.

2.7. Equivalent representations

The criterion is satisfied by structure, not by a particular schema. The typing and the relation may be carried as labels, classifications, taint tags, lineage metadata, correlation rules, proof circuits, policy graphs, or an ontology. The paper claims only that *some* representation of the typing and the relation must be present. A mechanism that answers “did the barrier hold” must represent the derivation relation and the typing of source and sink; a mechanism representing neither cannot answer it, whatever it is called. This is why the contribution is a criterion over representations—and, in Section 3, one specific representation shown to satisfy it—rather than a claim that only one representation will do.

3. Operationalising the Criterion: A Machine-Checkable Evidence Layer

The criterion is a specification. This section discharges the computational-law obligation the specification creates: to exhibit a legal knowledge-representation artefact that realises the two features and a rule set that checks them by machine. The artefacts are deposited, with a reproducibility pack, on Zenodo [26]; this paper reproduces the load-bearing fragments so the operationalisation can be read and criticised in place.

3.1. The two features as reused vocabularies

The criterion’s two features map cleanly onto two mature W3C vocabularies, which the ontology reuses rather than reinventing:

- *Typing* → *Data Privacy Vocabulary (DPV)* [4]. The legal typing of data, purpose, legal basis, and data subject is carried by DPV classes (`dpv:PersonalData`, `dpv:Purpose`, `dpv:LegalBasis`, `dpv:DataSubject`), extended by a

Table 1 . Necessity by construction. In each row the truth of D depends on a typing and a relation the bare log does not carry, so by Definition 1 it cannot answer D .

Determination D	Truth value fixed by	Why D is not recoverable from the bare log	Minimal addition
Did protected data leave the controlled environment?	Whether data of a protected class was the content that crossed an external boundary.	The log shows a send occurred and was permitted, not the class of the data nor the boundary type.	Typing of data by protected class; typing of channel as internal/external (DPV).
Was a human able to intervene?	Whether a human held authority and a usable window to override at the decision point.	The log shows an action executed, not whether an override path existed or was reachable in the available time.	Representation of authority state and intervention window at the decision point.
Did the information barrier hold?	Whether information derived from a high-side source reached a low-side sink.	The log records two permitted actions, not the derivation relation between them.	Provenance linking the send to the source read; typing of source and sink (PROV-O).
Was delegated authority valid when used?	Whether the authority was granted, unrevoked, and in scope at the instant of use.	The log records that the action occurred, not the validity state of the authority at that instant.	Temporal model of delegation and revocation; binding of the action to the authority state.

```
gov:AgentActionShape a sh:NodeShape ;
  sh:targetClass gov:Activity ;
  sh:property [ sh:path dpv:hasLegalBasis ; sh:minCount 1 ; sh:class dpv:LegalBasis ;
    sh:message "Activity contributing to a Decision must declare a dpv:hasLegalBasis under GDPR Art. 6(1)." ] ;
  sh:property [ sh:path dpv:hasPurpose ; sh:minCount 1 ; sh:class dpv:Purpose ;
    sh:message "Activity contributing to a Decision must declare a dpv:hasPurpose under GDPR Art. 5(1)(b)." ] .
```

Figure 1. The typing feature as a cardinality constraint: an activity that feeds a decision must carry a lawful basis (GDPR Art. 6(1)) and a purpose (Art. 5(1)(b)), typed through the Data Privacy Vocabulary, or the record cannot answer under what legal category the data was processed.

supervisory namespace `gov:` for AI Act constructs (risk classification, intended purpose, legal role) that DPV does not cover.

- *Relation* $\rightarrow$ *PROV-O* [3]. The provenance, derivation, and attribution relations are carried by *PROV-O* (`prov:Activity`, `prov:wasAssociatedWith`, `prov:wasGeneratedBy`, `prov:wasDerivedFrom`). Temporal validity, which *PROV-O* does not fully model, is carried by an explicit `gov:DelegationGrant` / `gov:RevocationEvent` construct with validity intervals.

The supervisory-evidence ontology organises these into 23 conceptual slots (an agent-behaviour core and a supervisory-evidence layer); under strict reuse-zero accounting, seven slots are DPV/PROV-O reuses and 16 are net-new classes. The design point is the correspondence: the criterion’s “typing” and “relation” are not abstractions awaiting a schema; they are already-standardised representational commitments, which is what makes a machine-checkable form available.

3.2. The criterion as SHACL: rules as code

The Shapes Constraint Language (SHACL) [2] lets the criterion’s conditions be expressed as constraints a validator enforces over an evidence graph. Three shapes illustrate the range, from a cardinality constraint realising the typing feature—lawful basis and purpose (Figure 1)—through a shape requiring a typed validity interval on delegated authority (Figure 2), to a SPARQL shape encoding case-law reasoning (Figure 3). The shapes are reproduced from the deposited profile `mcs_profiles_v1_2.ttl` [26], with `sh:message` strings abbreviated for space.

The activity shape (Figure 1) realises the typing feature: an activity feeding a decision must carry a lawful basis and a purpose. The delegation shape (Figure 2) realises the relation feature for a determination that turns on temporal validity the bare log omits. The

Profile B scope shape (Figure 3) is legal knowledge representation in the strict sense: the *SCHUFA* distinction,³ with the *Uber* intervention bar,⁴ rendered as an executable SPARQL constraint that surfaces a candidate misclassification for a human to adjudicate.

Dual profiles for legal divergence. Supervisory interpretations of Article 22 and of “meaningful human intervention” diverge across authorities and courts. Rather than resolve the divergence, the ontology encodes it: a permissive Profile A and a stricter Profile B (the latter requiring, e.g., `gov:authorityActuallyExercised` in line with the *Uber* judgment). A decision graph can be validated against both; the delta between conformance under A and under B becomes an explicit governance artefact rather than a hidden legal assumption. Representing legal indeterminacy as a checkable, versioned profile difference is itself a computational-law technique the criterion makes usable.

3.3. A worked validation

The criterion predicts that a bare or under-typed record fails to answer the determinations and a typed relational record answers them. Under SHACL this prediction becomes an executable pass/fail test. Consider a minimal deficient graph (Figure 4); validated against the profile, it does not conform, and the validator returns typed, legally annotated violations, each naming the missing typing or relation and the obligation it serves (Figure 5).

By contrast, a graph in which the same entities carry the required typing and relations—legal basis and purpose on the activity, a typed validity interval and authority scope on the delegation,

³Case C-634/21 *SCHUFA Holding (Scoring)*, judgment of 7 December 2023, ECLI:EU:C:2023:957.

⁴Gerechtshof Amsterdam, *Uber B.V.*, judgment of 4 April 2023, ECLI:NL:GHAMS:2023:793. The parallel *Ola* appeal is ECLI:NL:GHAMS:2023:804.

```

gov:DelegationGrantShape a sh:NodeShape ;
sh:targetClass gov:DelegationGrant ;
sh:property [ sh:path gov:hasValidityInterval ; sh:minCount 1 ; sh:class gov:ValidityInterval ;
  sh:message "Every DelegationGrant must declare a validity interval (gov:hasValidityInterval)." ] ;
sh:property [ sh:path gov:hasAuthorityScope ; sh:minCount 1 ; sh:class gov:AuthorityScope ;
  sh:message "Every DelegationGrant must declare an authority scope (gov:hasAuthorityScope)." ] .

```

Figure 2. The relation feature for delegated authority: a delegation must carry a typed validity interval and an authority scope, so that “was the authority valid, and in scope, when used” becomes checkable over the interval rather than assumed. A companion *RevocationEventShape* binds each revocation to the grant it revokes and to a timestamp.

```

gov:ProfileB_Art22ScopeShape a sh:NodeShape ;
sh:targetClass gov:Score ;
sh:sparql [
  sh:prefixes [ sh:declare [ sh:prefix "gov" ;
    sh:namespace "http://apparens.nl/mcs/gov#"^^xsd:anyURI ] ] ;
  sh:select """
    SELECT $this WHERE {
      $this a gov:Score .
      ?d gov:drawsStronglyOn $this ; gov:hasLegalEffect true .
      FILTER NOT EXISTS {
        ?d gov:hasHumanIntervention ?hi .
        ?hi gov:hasAuthorityToDeviate true ; gov:hasMarginOfDiscretion true ;
          gov:hasUnderstandingOfLogic true ; gov:consideredAllRelevantData true ;
          gov:authorityActuallyExercised true . } }
    """ ;
  sh:message "Profile B (CNIL-strict): Score engages Art. 22(1) via a downstream legal-effect decision drawing strongly on it,
    unless intervention meets all five conjunctive properties including gov:authorityActuallyExercised (Uber, ECLI:NL:GHAMS
    :2023:793)." ;
  sh:severity sh:Info ] .

```

Figure 3. Encoding case-law as a SPARQL constraint. In *SCHUFA Holding* (C-634/21) the Court of Justice held that an automated score can itself be an Article 22 GDPR decision where a downstream actor draws strongly on it and the decision produces legal effects. Profile B flags any such score *unless* a human intervention satisfies all five conjunctive properties—including *authorityActuallyExercised*, added in response to the Amsterdam Court of Appeal’s finding, in the *Uber* judgment (ECLI:NL:GHAMS:2023:793), that a human review amounted to no more than a “symbolic act” (the parallel *Ola* appeal is the separate judgment ECLI:NL:GHAMS:2023:804). The case-law distinction is thereby executable, and surfaces candidates for human legal judgement rather than adjudicating them.

and the six conjunctive intervention properties (authority to deviate, understanding of logic, consideration of relevant data, margin of discretion, intervention duration, authority actually exercised)—conforms. The move from the first graph to the second is exactly the move from a record that cannot answer the determination to one that can, and the SHACL validator is a concrete instance of the answering procedure f of Definition 1. The criterion is, in this precise sense, decidable against a declared profile.

4. Why Existing Governance Frameworks Do Not Answer These Determinations

A reviewer may suspect the criterion is already satisfied by a framework in wide use, or by an existing accountability ontology, so that the contribution is a relabelling. It is not. Frameworks that govern AI systems either produce records without the required structure or specify processes rather than evidence, and the semantic artefacts that do exist model the governance process rather than the runtime evidence graph (Table 2).

Process frameworks (NIST, ISO/IEC 42001) govern function, not evidence; record frameworks (Article 12, generic logging) produce data without typing and relation; components that carry one feature do not carry both. The semantic artefacts are the closest prior art and deserve a precise statement of the difference: they are representations of the *governance apparatus*—plans, roles, risks, rights, obligations—whereas the criterion concerns a representation of *what the system did*, from which a binary finding of fact about

a particular execution can be recovered by procedure. An organisation could hold a fully populated RAIInS accountability plan and an AIRO risk classification and still be unable to establish whether a given send carried protected data derived from a high-side read under a valid delegation. That gap is what the criterion names and what the operationalisation of Section 3 closes.

5. Theoretical Grounding

The criterion is argued from the determinations and operationalised in Section 3; it does not depend on any borrowed theorem. Three results from cybernetics and computer science nonetheless converge on the same necessity from different directions, and are recorded here as corroboration, not as the source of the criterion. The paper claims no novelty in the convergence.

- *Requisite variety* [19]: a point-in-time audit, taken as the sole oversight mechanism, acts once against a system that generates behavioural variety continuously, and so is in the wrong class to hold a runtime-dependent variable in range. A legal condition becomes such a variable only once it is made measurable and sensed through a channel; where the record cannot be read for the operative fact, it cannot be that channel.
- *The Good Regulator Theorem* [20]: a regulator coupled to a system in a feedback relation must carry a model of it. Applied to the overseer rather than the AI, this supports model-dependence—but the theorem locates a dependence; it

```

ex:activity_missing_basis a gov:Activity ; dpv:hasPurpose ex:purpose1 .
ex:decision_missing_effect a gov:Decision .
ex:delegation_missing_scope a gov:DelegationGrant ; gov:hasValidityInterval ex:notTypedInterval .
ex:humanReview_missing_exercise a gov:HumanIntervention .

```

Figure 4. A minimal deficient evidence graph: entities present, but missing the typing and relations the determinations depend on.

```

sh:conforms false ;
# delegation_missing_scope:
# "Every DelegationGrant must declare an authority scope (gov:hasAuthorityScope)."
```

Figure 5. The validation report for the deficient graph: sh:conforms false, with each violation naming the missing typing or relation and the GDPR/AI Act obligation or case-law source it serves.

does not supply the model's *legal* content, which comes from the criterion. The inversion onto the overseer is prior work⁵ and is disclaimed as novel here.

- *Runtime monitorability and hyperproperties* [21, 22]: Clarkson and Schneider show that important information-flow policies are hyperproperties rather than properties of a single trace—observational determinism is a 2-safety hyperproperty, and particular formulations of noninterference reduce to safety through self-composition—so such policies are not evidenceable from a single black-box trace. The honest reading is an instrumentation requirement, not an impossibility: gray-box monitoring lifts the limit when the monitor is given knowledge of the system [23], a point the auditing literature reaches independently in showing that black-box access bounds what an audit can conclude [14]. That the additional knowledge should consist specifically of the criterion's legal typing and truth-relevant relations is this paper's inference, not a conclusion of the monitoring literature.

If all three results were withdrawn, the criterion would stand on its construction (Section 2) and its operationalisation (Section 3); only the corroboration would be lost.

None of these literatures independently proves the criterion. Cybernetics establishes information and model constraints on effective regulation; computer-security research identifies policies whose truth depends on relations across executions; EU data-protection jurisprudence demonstrates that a legally meaningful characterisation may depend on the relation between a technical output and a downstream decision (Section 3); and semantic-web standards show that typing and provenance can be represented and structurally validated computationally. Together these literatures corroborate, but do not independently establish, the minimum evidentiary criterion developed here; the necessity claim rests on the construction of Section 2, and each borrowed result is used only for what it proves under its own stated assumptions.

⁵A. Aguirre, *Control Inversion: Why the Superintelligent AI Agents We Are Racing to Create Would Absorb Power, Not Grant It* (Future of Life Institute, 6 November 2025), <https://control-inversion.ai/>.

6. The EU-Law Instance

The criterion is domain-neutral; its legal force in this instance comes from the AI Act's obligations, split across Articles in conditional terms. Article 12 requires that a high-risk system technically allow automatic recording of events for traceability. Article 14 requires that natural persons can *effectively* oversee the system, including the ability to understand, intervene, and override. Article 26 places a monitoring and log-keeping duty on deployers; Article 72 requires post-market monitoring. In legal language, these describe a control loop: a duty to sense, a duty to be able to decide, and a duty to be able to intervene. The "control loop" reading is this paper's analytical synthesis of these provisions; the Regulation does not use the term, and it does not expressly require an ontological or semantic representation of the record.

The bearing of the criterion must be split across these Articles because they demand different things. Article 12 is a *recording* duty: an uninterpretable log may satisfy a substantial part of it, and the paper does not claim a bare trace fails Article 12. Article 14 is the stronger anchor: oversight that cannot answer the operative determination is not effective oversight. What Article 14 demands is contested—its aims, timing, and addressee are underdetermined on the face of the text [17], oversight duties are frequently unsatisfiable by the humans assigned to them [16], and "effectiveness" is itself an interdisciplinary construct rather than a formal one [18]. The criterion does not resolve that contest. It establishes a floor beneath it: on any reading of Article 14, oversight requires that the operative fact be recoverable from the record, and where it is not, the dispute about who oversees, when, and how well does not arise. A record that satisfies the letter of Article 12 recording can therefore still fail to support Article 14 oversight, and the gap between the two is exactly the typing and the relation the criterion identifies—and, in Section 3, exactly what the SHACL profile checks.

The claim is thus two directions of unequal strength. The *sufficiency* direction is weak and conceded: a structured supervisory record can serve as evidence input toward Article 14 and Article 26, but is not itself a discharge of those duties, which depend on facts about the deployment no representation can supply. The *necessity* direction is strong: a record that is not semantically and relationally interpretable cannot serve as evidence that effective oversight under

Table 2 . Existing frameworks against the answerability criterion. No framework in current use produces, by itself, a record satisfying the criterion—which is what the ontology and shapes of Section 3 are built to close.

Framework	Produces records / findings?	Gap against the criterion
NIST AI Risk Management Framework ISO/IEC 42001	Process and outcomes; no direct record duty. Management-system certification.	Governs organisational function, not an evidentiary representation that types events and carries relations. Certifies that a management system operates; does not define the content of a record from which a legal finding is recoverable.
EU AI Act, Article 12 (logging) Runtime verification / enforcement Provenance models (PROV and successors) Accountability ontologies (SAO / RAIInS) [8]	Automatic event recording over the lifetime. Enriched traces, as a by-product of mediation. Derivation graphs; the relation. Accountability plans and their traces, by design stage.	Does not, on its face, require logged events to be typed to legal categories or connected by the relations on which findings depend. A black-box single-path monitor cannot evidence relational properties without system knowledge (Section 5); audit strength tracks access [14]. Carry the relation but not the legal typing: they answer “was <i>Y</i> derived from <i>X</i> ” but not “was <i>X</i> of protected class <i>C</i> .” Model the governance process—plans, roles, obligations—not the runtime evidence graph; state no necessity condition on record content and no conformance test for a finding of fact.
AI Act risk vocabularies (AIRO / VAIR) [6] GDPR ontologies and policy languages [5]	Machine-readable risk and high-risk classification. Rights, obligations, informational items.	Type the <i>system</i> and its risks, not the events and relations of a particular execution; classification is not answerability. Represent what the law requires; do not represent the runtime relation on which a finding about a specific execution turns.

Article 14 was possible, because the operative fact is not a function of the record’s content. A doctrinal analysis of agents under EU law reaches a compatible conclusion, that high-risk agentic systems whose behaviour cannot be traced cannot satisfy the Act’s essential requirements [24]. The step from their “untraceable” to this paper’s “unanswerable” is our own inference, not a conclusion the preprint itself draws.

The GDPR jurisprudence relied on here—*SCHUFA* on when an upstream score is itself an automated decision, and *Dun & Bradstreet Austria* on the Article 15 requirement of an intelligible explanation of an automated decision⁶—demonstrates that a legally meaningful characterisation can depend on the relation between a technical output and a downstream decision. It is invoked as adjacent data-protection authority for that relational point, not as an interpretation of Article 14 of the AI Act, on whose human-oversight standard the Court of Justice has not yet ruled.

Timing, in conditional terms, perishable.. Under Article 113 of Regulation (EU) 2024/1689 as originally enacted, the high-risk obligations, including the Article 12 record-keeping and Article 14 human-oversight duties, become applicable on 2 August 2026 for standalone Annex III systems and 2 August 2027 for Annex I product-embedded systems. The Digital Omnibus on AI, adopted by the co-legislators and signed on 8 July 2026 following the Council’s final approval on 29 June 2026, defers these: it amends Article 113 to set 2 December 2027 for standalone Annex III high-risk systems and 2 August 2028 for Annex I product-embedded systems.⁷ As of 21 July 2026 the amending Regulation is adopted and signed but not yet published in the Official Journal: no Official Journal number was available at the time of writing, and it has not yet entered into force. The definitive text is therefore cited as the adopted consolidation PE-CONS 30/1/26. The mechanism of the deferral is material to this paper’s argument: the stated rationale is to align

the application date with the availability of the harmonised standards and support tools⁸—the very standards that would specify an oversightable record and do not yet exist. The eventual Official Journal reference—*Regulation (EU) 2026/...*—will change neither the application dates set out above nor this paper’s argument, whose substance is unaffected by the deferral.

7. Implementation and Integration

The operationalisation of Section 3 is designed to sit inside a supervisory evidence pipeline, not beside it. The intended data flow is:

1. *Emit*. Runtime events (tool invocations, memory reads/writes, decisions, human interventions) are emitted by the controller’s system as an RDF graph, natively or via an adapter over existing logs.
2. *Type*. Data, purpose, and legal-basis references are typed with DPV classes; AI Act constructs with the gov: extension—realising the criterion’s typing feature.
3. *Relate*. Activities are linked by PROV-O provenance, and delegated authority by validity-interval-bearing grant/revocation events—realising the relation feature.
4. *Validate*. The graph is validated against the SHACL profile (and, where interpretations diverge, against Profiles A and B). Conformance is the machine-checkable instance of Definition 1; non-conformance returns typed, legally annotated violations.
5. *Assemble*. The conformant graph plus its validation report forms a per-use-case accountability file a supervisor can ingest, compare across controllers, and query, without bespoke schema reconciliation.

This is where the contribution meets legal technology in practice: a controller keeps its internal operational schema and emits supervisory evidence in the shared form; a supervisory authority

⁶Case C-203/22 *Dun & Bradstreet Austria*, judgment of 27 February 2025, ECLI:EU:C:2025:117.

⁷Regulation of the European Parliament and of the Council amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI), adopted text PE-CONS 30/1/26 REV 1, procedure 2025/0359(COD), signed 8 July 2026.

⁸European Commission, *An agile Digital Rulebook for the EU*, <https://digital-strategy.ec.europa.eu/en/policies/digital-rulebook>.

ingests conformant graphs rather than narrative attestations; and a compliance function can run the profile in its own pipeline as a pre-submission check, at the point in the conformity-assessment and post-market-monitoring cycle where evidence is assembled for an external reviewer [15]. The pipeline is the machine-checkable component of a record-keeping design of the kind reviewability prescribes across the technical and organisational elements of a decision process [10]; it does not replace that design, and it covers only the runtime layer of it. The design states its own operating envelope: a pre-registered technical benchmark targets p95 validation latency below 100 ms at 100 decisions per second across at least pySHACL, Apache Jena SHACL, and the TopBraid SHACL API, so that “machine-checkable” is held to a throughput claim.

Two models, two predictions.. The criterion separates current oversight (a tamper-evident record of actions and decisions, read by a human against a checklist, with relational facts reconstructed by manual inference) from semantic oversight (a runtime record that types events and carries relations, from which a determination is recovered by procedure). Two falsifiable predictions follow. *Prediction 1:* organisations relying on action logs without typing and relation will systematically fail to answer the in-scope Article 14 determinations, while those whose records carry typing and relation will answer them for provenance-capturable flows. *Prediction 2:* increasing logging volume without adding typing and relation yields diminishing oversight value, because the determinations that remain unanswered are unanswerable in principle from records of that structure, not for want of volume.

8. Empirical Validation

The criterion is a necessity claim about whether a determination is recoverable from a record; such a claim is testable, and testing it is the difference between an argument and a result. It is also the defence against a triviality reading: if real deployed logging already carries the typing and the relation, the criterion is trivial; if it systematically does not, the criterion has bite. Two tracks test it.

Determination-answerability experiment (pre-registered).. A between-groups design with three independent expert panels avoids the learning-effect confound of a within-subjects sequence. Each panel receives the same cases, drawn from a 25-case EU automated-decision-making enforcement sample, in one of three record conditions: Condition A, an action log only; Condition B, the log plus a provenance graph (relation, no legal typing); Condition C, the typed relational representation realised through the ontology and SHACL profiles. Each case poses the four in-scope determinations. The primary measure is the proportion resolved correctly against a two-coder ground truth (Cohen’s κ reported); “cannot determine from this record” is an explicit, encouraged response, because conflating answerability failure with a guess would mask the effect. *Falsification:* the criterion is false or trivial if Condition A resolves the relational determinations at a rate indistinguishable from Condition C; the minimality claim is challenged if Condition B matches Condition C on the typing-dependent determination. Both outcomes are possible under the design, which is what makes it a test.

Technical feasibility benchmark.. The SHACL validation is benchmarked for throughput as described in Section 7, establishing that the machine-checkable form is deployable at supervisory scale and separating the representational claim from any performance objection.

9. Critical Evaluation and Limits

An argument of this kind is strengthened by stating where its strongest attacks land.

Necessity, not sufficiency. A record that satisfies the criterion, and conforms to the SHACL profile, can still fail to support good oversight: it can be curated to show compliance and omit what would reveal a breach, in the manner Power describes for performative risk management [25]. SHACL conformance buys legibility, not truthfulness; it makes a record readable for a determination without making it honest. The criterion relocates the failure from a record the supervisor cannot read at all to a record the supervisor can see was curated—auditable in a way the first is not—but it does not close it.

Structural completeness is not legal compliance. A conformant graph is a claim of structural completeness against a declared profile, not a proof of compliance. Legal classification remains a human act; the shapes surface candidates (as the *SCHUFA* shape surfaces a score for review) and check structure, they do not adjudicate.

A relation is not yet an inference. The criterion preserves relations of three kinds that the analysis of proof keeps distinct [1]: *world* relations (what happened, and between whom—*wasGeneratedBy*, *actedOnBehalfOf*, temporal precedence), *evidentiary* relations (why one recorded fact supports a further proposition), and *legal* relations (why that proposition satisfies an operative legal element). PROV-O and the typing vocabularies represent the first kind and make the second and third *formulable*; they do not supply them. That a provenance edge links a high-side read to a low-side send does not, on its own, warrant the inference that the read determined the send—that step requires a warrant the criterion does not encode. The claim is therefore deliberately modest: the criterion preserves the facts and relations from which an evidentiary argument can later be constructed, and stops short of the argument itself, its assessment of credibility, and the applicable standard of proof.

Bounded coverage. The criterion closes the gap only for flows crossing mediated, provenance-capturable surfaces. Information moving through model-internal latent state, opaque tool execution, summarisation that destroys provenance, or human transfer between screens is outside it. This is a real limitation, stated as one, not minimised; extending the boundary is an instrumentation problem the criterion presupposes and does not solve.

Triviality and novelty, as tests. If deployed records already carry these features, the experiment will show Condition A answering the determinations and the criterion is trivial; the paper exposes this rather than assuming non-triviality. The cybernetic inversion is not novel and is attributed to prior work (Section 5); the contribution is located in the criterion, its executable operationalisation, the gap analysis, and the test.

10. Implications and Generalisation

For boards and accountable functions, the distinction the criterion names is between integrity and answerability: a tamper-evident log can show a record was not altered and still fail to show what legally operative fact it establishes. For the operationalisation of the AI Act, the criterion names what the harmonised standards presuppose and do not yet specify: the typing and the relation are the part a standard for oversightable logging must fix for an Article 12 record to serve Article 14. For the design of oversight, the implication is a closed loop with a sensing channel that answers the determination, not a periodic audit.

The structure is not specific to AI. Wherever a legal finding about an autonomous system depends on its execution path, on the relation between actions, or on the provenance of a flow, the criterion applies and the same failure recurs: a record can have integrity and still fail to answer the finding. Autonomous trading systems must answer whether a barrier held or an authority was valid; medical systems whether a contraindication was visible at the decision point; critical-infrastructure controllers whether a command derived from an authenticated source. Each is a finding of fact about events and relations, in scope by the selection rule, and each is unanswerable from a bare event log for the same structural reason. EU AI Act oversight is the worked instance because its obligations are explicit and current, and the criterion is stated so the instance can be lifted.

11. Conclusion

For a defined class of legal determinations—findings of fact about the execution paths of autonomous systems—a runtime record is evidence of the finding only when it carries the typing that maps events to the operative legal category and the relation on which the finding's truth depends. The condition is a necessity claim, shown by construction, minimal, and bounded to the class. In the vocabulary of the analysis of legal proof, the two features are the minimum representational substrate from which an evidentiary argument connecting a runtime record to a legally material probandum can be constructed; they are not that argument, and they do not measure its probative force [1]. Its computational-law content is that this condition is not merely conceptual: it corresponds to reuse of DPV and PROV-O over a supervisory-evidence ontology, and it is rendered machine-checkable as SHACL shapes, including a temporal validity-interval shape on delegated authority and a SPARQL shape encoding the Court of Justice's *SCHUFA* reasoning, with a worked pass/fail validation and a decidable conformance test standing in for the answering procedure. Instantiated in the AI Act, the criterion yields a conditional legal claim—a record that is not semantically and relationally interpretable cannot evidence that effective Article 14 oversight was possible, even where it satisfies the Article 12 duty to log—and a gap analysis showing that no framework in current use produces such a record by itself. The criterion is testable, and the paper pre-registers the test rather than assuming its outcome.

A. Claim and Falsification Register (condensed)

The register decomposes the argument into load-bearing claims, grades each by evidential strength (A canonical/established; B well-supported; C defensible-but-contestable), and states a falsifier for the most exposed (Table 3). Only the falsifier is hunted; searching for support is excluded by design.

Falsifiers for the exposed critical claims. K4/K6: an in-scope determination that a bare, untyped, provenance-free log answers on its own. K7: a demonstration that one of the two features is redundant. K8: an in-scope determination unanswerable even with “relation” read to carry provenance, authority, and temporal structure. K11: a framework in current use, or an existing accountability ontology, that by itself produces a record typing events to legal categories and carrying the relations on which findings depend. K12: a determination in the class whose answering condition cannot be expressed as a decidable constraint over the evidence graph. K23/K24: a legal reading on which Article 12 already requires semantic interpretability, or on which a non-interpretable record evidences that effective

oversight was possible. K29: Condition A answering the relational determinations at the rate of Condition C.

B. Experiment Protocol (pre-registered)

Hypothesis. For determinations depending on a relation (information barrier; intervention window), the proportion resolved correctly is ordered $C > B > A$, with A at or near floor. For the determination depending on typing but not relation (protected-class boundary crossing on a directly typed channel), the order is $C > A$ with B near A, isolating the typing feature.

Design. Between-groups, three independent expert panels, one condition each. **Materials.** The 25-case EU automated-decision-making enforcement sample, each case rendered in Conditions A (action log), B (log + provenance graph), and C (typed relational representation via the ontology vocabulary and SHACL profiles). **Ground truth.** Two independent coders per case, disagreements adjudicated, Cohen's κ reported. **Participants.** Practitioners with expertise in data protection, AI assurance, or audit; panel size fixed by a power analysis targeting the predicted between-group difference on the relational determinations, run before recruitment; assignment randomised. **Procedure.** Each participant receives only their condition's cases; per determination, one of {yes, no, cannot determine from this record}, plus time-to-resolve and ordinal confidence. **Measures.** Primary: proportion correct against ground truth, by condition and determination type. Secondary: “cannot determine” rate, median time, within-panel agreement (Fleiss κ), confidence. **Falsification.** As in Section 8. **Status.** Design and materials fixed; recruitment, power analysis, and run pending. The argument does not presuppose the outcome.

Ethical Statement

This paper reports no completed study involving human participants. The determination-answerability experiment in Appendix B is pre-registered but not yet conducted; it involves professional expert coders assessing document artefacts, not personal data of data subjects, and will be submitted for the applicable ethics review and informed-consent procedures before any data collection.

Conflicts of Interest

The author develops the artefacts described here through Apparens, which provides AI governance services and therefore has a professional interest in the adoption of specifications of this class. For that reason the paper treats distributive effects, SME proportionality, and the distinction between structural validation and legal compliance as first-order design constraints. Generative AI tools were used for adversarial review, editing, and formatting; the author directed, checked, and approved the claims, citations, and conclusions and remains responsible for the content. The author declares no other conflicts of interest.

Table 3 . Condensed claim and falsification register.

ID	Claim	Load	Grade
K1	An agentic system selects its execution path at run time; the path is not enumerable in advance and can be steered by content the agent reads.	Critical	A
K2	E answers D iff D 's truth value is a function of E 's content (Definition 1).	Critical	A
K4	For an in-scope D , E answers D only if it carries the legal typing and the relation on which D 's truth depends (the criterion).	Critical	B
K6	Necessity holds by construction for the four determinations; the bare log makes none recoverable.	Critical	B
K7	The set {typing, relation} is minimal: dropping either loses some in-scope determination.	Critical	B
K8	The set is adequate for the class where "relation" carries provenance, authority, and temporal structure (bounded, not universal).	Critical	C
K9	Sufficiency fails: a conformant record can still be curated and is not thereby honest.	Critical	B
K10	The criterion is satisfied by structure, not a unique schema; DPV+PROV-O+SHACL is one satisfying representation.	Critical	B
K11	No governance framework in current use, and no existing accountability or AI Act ontology, produces by itself a record satisfying the criterion.	Critical	C
K12	The criterion's typing/relation conditions are expressible as SHACL constraints, including SPARQL and case-law-encoding shapes, and are decidable against a declared profile.	Critical	B
K22	AI Act Article 12 requires recording; Article 14 requires effective human oversight (understanding, intervention, override).	Critical	A
K23	A bare log may satisfy Article 12 recording while failing Article 14 oversight.	Critical	C
K24	A non-interpretable record cannot evidence that effective Article 14 oversight was possible (necessity); a structured record is evidence input, not a discharge (sufficiency, weak).	Critical	C
K27	Coverage is bounded to provenance-capturable surfaces; model-internal, opaque, summarisation, and human-transfer channels are outside it.	Critical	C
K29	The between-groups experiment discriminates the criterion from triviality; Condition A is predicted to fail the relational determinations.	Critical	C
K30	The ontology + SHACL deposit is an existence proof supplying the primitives (PROV-O, DPV, SHACL), not a validated reference architecture.	Supporting	A

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